

Composio Research Agent

App Integration Buildability Study

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Live page: <https://itzrahuldas.github.io/composio-research-agent/>

A 2-layer agentic research pipeline that analyzed 100 apps across 10 categories to identify buildability status for Composio tool integrations.

100

apps researched

66

ready-to-build today

93.3%

field verification
accuracy after
critique + repair

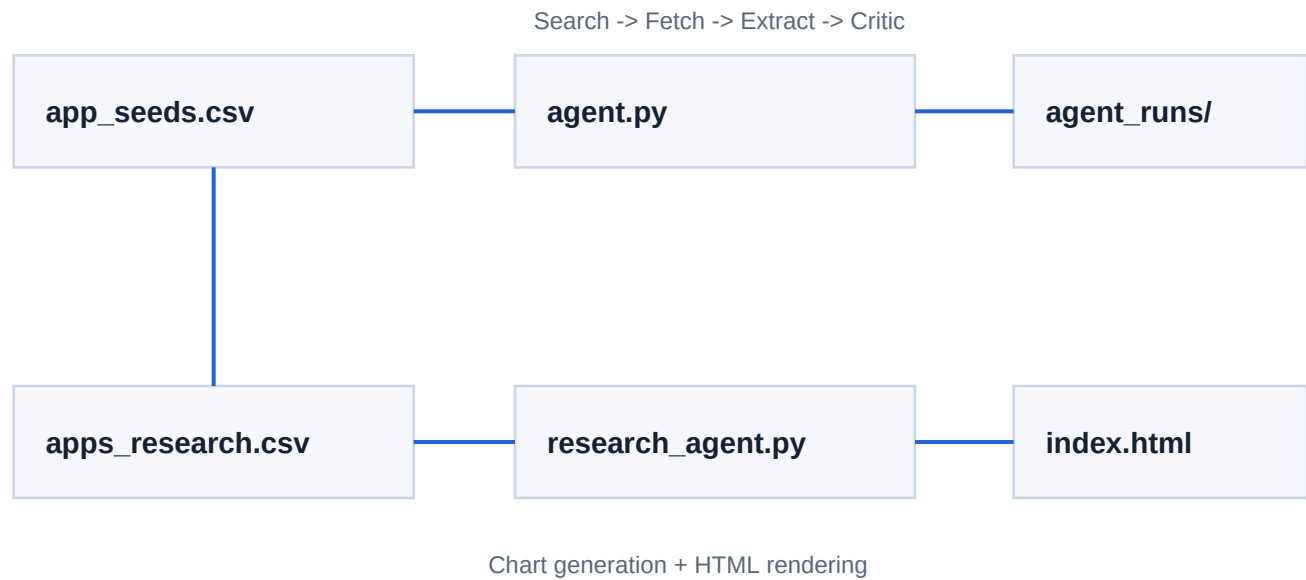
20

app live agent run
with critic flags

Primary outcome

- Build self-serve SaaS and developer platforms first.
- Queue review-gated ads, fintech and enterprise commerce next.
- Route hidden/private data products to outreach instead of pretending they are buildable.

2-layer system



Key components

- `src/agent.py`: web search, HTML parsing, heuristic extraction and critic flags.
- `src/research_agent.py`: computes patterns, renders charts, verification tables and the live HTML page.
- Composio SDK hook: optional `--use-composio` path for tool-mediated search when `COMPOSIO_API_KEY` is set.
- OpenAI LLM path: optional `--use-llm` path for structured extraction when `OPENAI_API_KEY` is set.
- Final matrix: human-repaired, evidence-backed rows in `data/apps_research.csv`.

Verdict breakdown

Status	Count
Ready today	66
Ready after gate	20
Outreach needed	9
No public build path	3
Investigate further	2

Top strategic insights

- Where to build first: Productivity (10/10), Dev/Infra (9/10), CRM (9/10) have clear docs and reusable OAuth/token patterns.
- Where outreach matters: Fintech and Ecommerce have the densest blocked/gated rows; this is licensing/access friction, not pure engineering difficulty.
- MCP is a wedge: 19 apps show official MCP or agent-native signals, so Composio can wrap existing MCP surfaces first.
- Auth x buildability: OAuth rows are about 92% buildable now/after normal gates; API-key rows are about 93%.

Human-in-the-loop process

- First pass: live agent ran on 5-20 app samples per batch.
- 27 apps manually sampled, stratified across all 10 categories.
- 135 field checks performed across auth, access, surface, verdict and evidence.
- First pass: 90/135 fields supported (66.7%).
- After critique + repair: 126/135 fields supported (93.3%).
- Misses are documented in `verification_field_audit.csv` with evidence URLs.

Honest limitation

DuckDuckGo rate-limits and many enterprise doc pages are JS-rendered, making raw HTML scraping noisy beyond roughly 20 apps. The final matrix combines agent extraction with human verification, which is a realistic production pattern for high-accuracy API research.

```
# Step 1: Install
pip install -r requirements.txt

# Step 2: Run live agent
python src/agent.py --limit 20 --max-pages 3

# Step 3: Build output
python src/research_agent.py --build

# Optional: With LLM extraction
OPENAI_API_KEY=... python src/agent.py --use-llm --limit 5

# Optional: With Composio
COMPOSIO_API_KEY=... python src/agent.py --use-composio --limit 5

# Tests
python -m unittest tests/test_agent.py
```

Links

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